



In addition to part I (General handout for all courses appended to the timetable) this portion gives further details regarding the course.

*Course Number* : PHY F211

*Course Title* : Classical Mechanics

*Instructor-in-Charge* : Biswanath Layek (layek@pilani.bits-pilani.ac.in; 3242-B)

**Course Description:** This course introduces the principles of classical mechanics from Newtonian and analytical perspectives. It covers constrained motion, Lagrangian and Hamiltonian formulations, variational principles, central-force dynamics, canonical transformations, Poisson brackets, the kinematics and dynamics of rigid bodies in inertial, and rotating reference frames.

**Scope & Objectives:** To provide a strong foundation in analytical mechanics and equip students with the mathematical and physical tools required to formulate, analyze, and solve problems in classical mechanics with elegance and rigor. Importantly, the course serves as a gateway to advanced topics in theoretical physics, including quantum mechanics, statistical mechanics, electrodynamics, and general relativity.

**Course learning outcomes:**

- **Apply** vector calculus, generalized coordinates, and analytical methods to formulate and solve problems involving the motion of particles and systems with constraints using Lagrangian and Hamiltonian formulations.
- **Analyze** central-force motion and variational principles to derive equations of motion, identify conserved quantities, investigate orbital dynamics and stability, and explain the role of symmetries in classical mechanics.
- **Employ** Hamiltonian mechanics, canonical transformations, generating functions, and Poisson brackets to transform, simplify, and solve dynamical systems while recognizing canonical invariants.
- **Evaluate** the kinematics and dynamics of rigid bodies using orthogonal transformations, Euler angles, and rotating reference frames, including the analysis of Coriolis and related (non) inertial effects.

**Text Books:**

- T1.** H. Goldstein, C. Poole, and J. Safko, *Classical Mechanics*, Third Edition, Pearson Edu. Inc., 2002.  
**T2.** N. C. Rana and P. S. Joag, *Classical Mechanics*, McGraw Hill, First Edition, 2008.

**Reference Books:**

- L. D. Landau and E. M. Lifshitz, *Mechanics*, Third Edition, Butterworth-Heinemann, 2000.
- L. N. Hand and J. D. Finch, *Analytical Mechanics*, Cambridge Univ. Press, 1998.



**Course Plan:** T1 ==> Text Book-1, T2 ==> Text Book-2, CN ==> Class Notes

| Lecture | Learning Objectives                                                                       | Topics                                                                                                                                                                                                     | Learning Outcome                                                                                                                                                   | Ref.                                       |
|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 0       | A Few mathematical tools                                                                  | Vector Algebra: Basic Operations, Differential Operators (Gradient, divergence, curl, and Laplacian) in curvilinear coordinate systems, Multivariable Calculus (total derivatives and chain rule)          | Apply vector algebra, differential operators in curvilinear coordinate systems, and multivariable calculus to formulate and solve problems in classical mechanics. | CN (Self revision)                         |
| 1       | Kinematics & dynamics of a mechanical system in Newtonian approach: A very brief overview | Newtonian mechanics for a particle/system of particles: Conservation principles, etc.                                                                                                                      | Recall the fundamental concepts of Newtonian kinematics and dynamics relevant to analytical mechanics.                                                             | T1: 1.1 - 1.2 (Self revision), CN          |
| 2-3     | Newtonian approach: A Critical analysis                                                   | Constrained Motions: Classification of constraints. Degree of freedom (DOF), <i>generalized</i> coordinates.                                                                                               | Determine the degrees of freedom of mechanical systems, and formulate their motion using generalized coordinates.                                                  | T1: 1.3<br>T2: 1.1-1.2, 2.2, 2.3 and CN    |
| 4-6     | Lagrangian Formulation ( <i>user-friendly packaging</i> ):                                | Systematic methods to eliminate/bypass constraints, Principle of virtual work, d'Alembert's principle ( <i>may shock you at first and certainly test your patience!</i> ), Euler-Lagrange's (E-L) equation | Learning of alternate approach but more elegant, and most importantly, powerful, compared to Newtonian.                                                            | T1: 1.4-1.6<br>T2: 1.3, 1.4, 1.7, 2.1, 2.4 |
| 7-8     | Central force problem: A detailed study                                                   | Setting up the problem through Lagrangian approach: Reduction of two-body problem to an equivalent one-body problem.                                                                                       | Apply the Lagrangian formalism to analyse central-force motion and derive its equations of motion.                                                                 | T1: 3.1                                    |
| 9-12    | Qualitative understanding of power law forces: An <i>Effective Potential</i> description  | Two first integrals of motion, classification of orbits based on bounded/unbounded motion, stability criteria, closure property, etc.                                                                      | Use conservation laws to analyse bounded and unbounded central-force orbits and determine their stability and closure characteristics.                             | T1: 3.2 – 3.3, 3.5-3.6                     |



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|-------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 13-14 | Kepler problems:<br>Study to our solar system                                                                       | Orbits under inverse-square law of force, any deviation from inverse square law? Qualitative understanding of precession of orbits via phenomenological model.                    | Analyse orbital motion under the inverse-square law and explain orbital precession arising from deviations from the inverse-square force.  | T1: 3.7-3.8                      |
| 15-17 | Calculus of variations (an <i>integral principle</i> approach) and its application to different branches of physics | Principle of <i>least action</i> : Fermat's principle in optics, Geodesics for material bodies, Material bodies in conservative force fields (Brachistochrone, Tautochrone, etc.) | Apply the principle of least action to analyse the motion of light and material bodies and solve elementary variational problems.          | T2: 6.1-6.4, 7.1-7.3             |
| 18-20 | Hamilton's principle: An elegant & more powerful approach                                                           | Derivation of the Euler–Lagrange Equations from Hamilton's principle                                                                                                              | Establish the connection with D'Alembert's principle.                                                                                      | T1: 2.1-2.5                      |
| 21    | Symmetry principle and conservation laws                                                                            | Derivation of conserved quantities from the symmetries of the Lagrangian                                                                                                          | Relate symmetries to conservation laws                                                                                                     | T1: 2.6-2.7                      |
| 22-24 | Hamiltonian Formulation: A Phase space description                                                                  | Construction of the Hamiltonian from Lagrangian ( <i>the Lagrangian's twin sibling!</i> ) through <i>duality</i> transformation.                                                  | Construct the Hamiltonian from the Lagrangian using the Legendre transformation and interpret the dual formulation of classical mechanics. | T1: 8.1, 8.2<br>T2: 5.1-5.3, 5.5 |
| 25-27 | Canonical Transformation(CT):                                                                                       | Four basic CTs and the corresponding generating functions (GF)                                                                                                                    | Analyse canonical transformations using generating functions                                                                               | T1: 9.1-9.4<br>T2: 8.1-8.5       |
| 28-29 | Poisson bracket(PB)                                                                                                 | Poisson bracket and other canonical invariants (where you <i>start sensing quantum mechanics lurking in the background!</i> )                                                     | Apply PB and canonical invariants to analyse dynamical systems                                                                             | T1: 9.5 and CN                   |



|       |                                     |                                                                                                                                    |                                                                                                                               |                               |
|-------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 30-32 | Kinematics of Rigid Body Motion     | Independent Coordinates, Orthogonal Transformations, Formal properties of transformation matrix                                    | Analyse rigid-body kinematics using orthogonal transformations, Euler angles.                                                 | T1: 4.1-4.3                   |
| 33-35 | Kinematics of Rigid Body Motion     | Euler Angles, Finite and Infinitesimal Rotations                                                                                   | Describe rigid-body orientation using Euler angles and analyse finite and infinitesimal rotations in three-dimensional space. | T1: 4.4, 4.7, 4.8             |
| 36-39 | Motion in Rotating Reference Frames | Rate of Change of a vector, Coriolis Effects ( <i>hard to avoid when you're sitting in a lecture room on the rotating Earth!</i> ) | Related to the realistic and physical aspects of earth's rotation, namely Coriolis effects.                                   | T1: 4.9, 4.10<br>T2: 3.1, 3.3 |

***Evaluation Scheme (Total Marks : 200)***

| No. | Evaluation Component                                                                                                       | Duration | Total Marks (percent) | Date & Time                                                                             | Nature                  |
|-----|----------------------------------------------------------------------------------------------------------------------------|----------|-----------------------|-----------------------------------------------------------------------------------------|-------------------------|
| 1.  | <b>Four</b> tutorial tests ( <b>16 marks</b> each) will be conducted, and the <b>best three</b> scores will be considered. | 30 mins  | 48 (24%)              | To be announced in class/Google Classroom at least one week prior to the scheduled test | Closed Book             |
| 2.  | <b>Two</b> take-home assignments (6 marks each)                                                                            | One Week | 12 (6%)               | The details will be shared in class/Google Classroom                                    | Open Book               |
| 3.  | MidSem Exam                                                                                                                | 90 mins  | 60 (30%)              | 08/10 (AN1), as per the AUGSD timetable                                                 | Closed Book             |
| 4.  | Compre-Exam                                                                                                                | 180 mins | 80 (40%)              | 12/12 (FN), as per the AUGSD timetable                                                  | Closed Book + Open Book |



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**Chamber Consultation Hour:** To be announced in class/Google Classroom

**Notices:** Notices will be displayed in **Google Classroom** only.

**Make-up Policy:** *No make-up for tutorial tests.* For midterm/comprehensive exam, make up will be granted to genuine cases after proper verification (case-by-case basis).

**Criterion for NC:** An NC grade shall be awarded if a student does not participate in any evaluation component or secures less than 15% of the course total.

**Attendance policy:** Each student is expected to attend all classes and participate in all evaluation components.

*Instructor-in-Charge*

*Biswanath Layek*